WEEKLY PROGRESS REPORT

TaxWise Freelance – Automated Tax Deductions & Invoicing Platform

Week 1 — Problem Discovery & User Personas

Project ID	PTUT-PRJ-042
Project title	TaxWise Freelance – Automated Tax Deductions & Invoicing Platform
Domain	5. FinTech, Personal Wealth & Micro-Finance
Reporting week	Week 1 of 12 — Problem Discovery & User Personas
Report date	21 October 2026 (period: 14–20 Oct 2026)
Repository	github.com/SohaibRamzan01/taxwise-freelance

1. Week 1 objectives

Required deliverable	Status
Interview target users to identify operational friction	Completed — interview protocol + 5 participant profiles, coded into 5 friction themes
2 detailed user personas (roles and pain points)	Completed — Maya (independent designer), Daniel (independent consultant)
1-paragraph problem statement	Completed
5 core MVP features	Completed — each with Given/When/Then acceptance criteria

2. Work completed this week

- Defined the target population: US freelancers, independent contractors and remote workers with self-employment income. Tax professionals are secondary stakeholders.
- Designed a 30-minute semi-structured interview protocol (consent, expense walkthrough, invoice-to-payment walkthrough, quarterly-tax planning, concept check) with ethics rules: no SSNs, bank credentials or real tax returns collected; notes anonymised.
- Built five representative participant profiles (designer, software contractor, delivery contractor, consultant, tutor) and coded observations into friction themes: capture delay, classification uncertainty, duplicate entry, estimate uncertainty and collection delay.
- Synthesised two personas, wrote the problem statement and fixed the MVP scope to five core features.
- Set up the project repository and orchestration workflow (task briefs, implementation reports) for the rest of the semester.

3. Key findings

Friction theme	Consequence for the user	Product response
Invoices and income kept in separate ledgers	Missing or duplicated income distorts profit	Marking an invoice *paid* creates exactly one linked income entry
Deductible amount confused with purchase price	Deductions are overstated	Store business-use % and show the deduction formula
Records pile up before they are categorised	Clean-up competes with billable work	Fast entry, filters, optional AI category suggestion
Payment history separate from tax forecasts	Users under- or over-pay quarterly estimates	Estimator with line-by-line breakdown, logged payments, remaining balance
Unpaid invoices mistaken for available income	Poor cash planning	Dashboard separates outstanding invoices from received income

4. Problem statement

> Freelancers and independent contractors who manage expenses, invoices, income and quarterly tax estimates in separate tools struggle to keep financial records consistent and timely. Duplicate payment entry, uncertain business-use deductions and disconnected payment histories make it difficult to understand net earnings and plan estimated federal taxes. TaxWise Freelance will provide a single tenant-isolated workflow for recording business activity, issuing invoices, calculating transparent planning estimates and monitoring payment deadlines, with explicit assumptions and an estimates-only disclaimer. The MVP does not file returns, move money, connect bank accounts or replace professional tax advice.

5. Five core MVP features

1. **Expense tracking & deduction categorisation** — 15 Schedule C-aligned categories; deductible = amount × business-use % × category factor (Meals 50%).
2. **Income tracking** — manual entries plus automatic income when an invoice is marked paid.
3. **Quarterly tax estimator** — SE tax, QBI, standard deduction, bracket tax, safe-harbour option, payments made, next due date.
4. **Invoicing** — clients, line items, per-user numbering, DRAFT/SENT/PAID/OVERDUE/VOID statuses, printable invoice.
5. **Dashboard** — YTD KPIs, income-vs-expense and deductions-by-category charts, outstanding invoices, upcoming deadlines.

6. Challenges and how they were handled

Challenge	Resolution
Tax rules are complex and change yearly	Scope fixed to US federal 2026 rules, stored as versioned data so the year can be swapped
Risk of the app being read as tax advice	Every estimate screen carries an "estimates for planning, not tax advice" disclaimer
Interview findings come from a small sample	Findings treated as hypotheses to validate with the prototype in Week 7

7. Plan for Week 2

- Write the formal IEEE 830 Software Requirements Specification.
- Write formal use-case specifications and draw the system use-case diagram.
- Specify quantifiable non-functional requirements (latency, security, uptime).

Appendix A — Full Week 1 deliverable

Week 1 — problem discovery

1. Research purpose and method

TaxWise Freelance studies how freelancers connect receipts, client payments, deductible expenses, and quarterly estimated taxes. The research should establish whether one workflow reduces duplicate entry and makes estimates understandable. The target population is US freelancers, independent contractors, and remote workers with self-employment income; tax professionals are contextual stakeholders, not the primary operators.

Interview plan

Recruit five participants across creative, technical, consulting, delivery, and tutoring work. Include different income regularity, expense volume, and spreadsheet experience. Recruitment should avoid collecting Social Security numbers, bank credentials, real invoices, or tax returns. Obtain consent, offer withdrawal, anonymise notes, and use redacted examples. A student researcher conducts a 30-minute session with a note-taker; do not record without explicit consent.

Time	Activity	Questions / evidence
0–5 min	Context and consent	What work do you do? How variable is income? Who handles bookkeeping?
5–13 min	Recent expense walkthrough	Describe the last business purchase. Where was it recorded? How was mixed personal/business use handled?
13–20 min	Invoice-to-payment walkthrough	Show a redacted or fictional invoice process. When does a payment enter income records? What gets entered twice?
20–26 min	Quarterly planning	How did you decide the last estimated payment? Where did the date and prior-year figures come from?
26–30 min	Concept check	Which dashboard number would you verify first? Would a suggested category require confirmation?

Code observations as capture delay, classification uncertainty, duplicate entry, estimate uncertainty, or collection delay. Separate observed behaviour from opinions. A later prototype study should time expense entry and invoice payment recording, and ask participants to explain an estimate breakdown. Five interviews explore workflow patterns; they cannot estimate population prevalence.

2. Five illustrative interview summaries

Evidence status: The following are representative fictional interviews created for requirements discovery. No interviews were conducted, and the quotes are invented. Counts below describe only these five scenarios.

ID / representative participant	Current workflow	Illustrative finding	Design implication
I-01 / independent designer, monthly retainers	Spreadsheet expenses, receipt folder, document invoices	Enters a paid invoice in a second sheet and sometimes forgets the entry. “Payment should finish the record.”	Marking an invoice paid creates exactly one linked income entry.
I-02 / software contractor, irregular contracts	Bank export, spreadsheet, recurring software receipts	Cannot distinguish a software suggestion from a confirmed deduction. “Show me why that amount counts.”	Keep category, business-use percentage, and deductible amount visible; AI remains optional.
I-03 / delivery contractor, many small purchases	Phone notes and paper receipts	Records mixed-use purchases at full value and postpones cleanup. “I need to enter the business share while I remember it.”	Expense entry captures business use and category-specific deductibility. Mileage calculation is outside this MVP.
I-04 / independent consultant, larger invoices	Invoice template plus calendar reminders	Has prior-year tax totals but no single view of payments already made. “I do not know whether I am catching up or paying twice.”	Compare annualised liability and safe-harbour target; subtract logged payments visibly.
I-05 / part-time tutor, seasonal income	Cash notebook and occasional spreadsheet	Treats a quiet month as representative of the year and misses a deadline. “A date and a breakdown would help more than one big number.”	Label annualisation assumptions; display the next due date and payment history.

Operational friction findings

Finding	Scenario evidence	Operational consequence	Requirement response
Separate invoice and income ledgers	I-01, I-04	Missing or duplicated receipts distort profit	One atomic paid-invoice/income operation (FR-03, FR-05)
Deductibility is confused with purchase price	I-02, I-03	Users overstate deductions	Store business share and show deduction formula (FR-02)
Records accumulate before classification	I-02, I-03	Cleanup competes with billable work	Fast entry, filters, optional suggestion/fallback (FR-09, FR-10)
Payment history and forecasts are separate	I-04, I-05	Payment amounts are hard to reconcile	Estimator breakdown and logged payments (FR-04, FR-06)
Cash timing obscures performance	I-01, I-05	Outstanding invoices look like available income	Separate outstanding invoices from received income (FR-07)

These are hypotheses for validation, not measured product benefits. A real study should check whether participants can explain deductible amounts and reconcile the dashboard with sample records.

3. User personas

Persona A — Maya, independent designer

- **Role and context:** Sole proprietor serving several small clients from a home workspace; comfortable with web tools, limited bookkeeping knowledge. Fictional persona synthesised from I-01 and I-02.
- **Goals:** Send consistent invoices, see received income, capture subscriptions and supplies promptly, and understand which part of a purchase is deductible.
- **Pain points:** Re-entering payments, forgotten recurring expenses, unclear mixed-use treatment, and a spreadsheet that lags behind client work.
- **Tools today:** Spreadsheet, document invoice template, email, cloud receipt folder, and personal calendar.
- **Illustrative quote:** “When the client pays, I want my income and dashboard to agree.”
- **Detailed scenario:** Maya adds a client, drafts two design line items, checks the total, then marks the invoice sent. A week later she records payment. One linked income entry appears using the payment date. She adds a $100 mixed-use subscription with 80% business use and sees an $80 deduction. The dashboard separates the received amount from an unpaid invoice. She can print the invoice through the browser's Save as PDF workflow.
- **Success evidence:** Recording payment once changes income once; she can explain the displayed $80 without opening a separate sheet.

Persona B — Daniel, independent consultant

- **Role and context:** Established consultant with uneven project receipts and prior-year tax records; uses desktop for planning and phone for capture. Fictional persona synthesised from I-04 and I-05.
- **Goals:** Plan quarterly cash reserves, compare a forecast with safe harbour, record payments already made, and identify overdue client balances.
- **Pain points:** Forecasts fluctuate after large receipts, payment history is fragmented, and a single tax total hides the self-employment component.
- **Tools today:** Spreadsheet forecast, calendar, invoice documents, and accountant-prepared prior-year summary.
- **Illustrative quote:** “Tell me the assumptions before I trust the payment amount.”
- **Detailed scenario:** Daniel selects the 2026 tax year and single filing status, enters prior-year total tax and AGI, then opens the estimator. It shows YTD profit, annualisation period, SE tax, deductions, federal income tax, previous payments, and the next scheduled date. He selects safe harbour, compares its target with the forecast, and records an estimated payment. The remaining balance updates while the calculation disclaimer stays visible.
- **Success evidence:** He can trace the target to the selected method and reconcile the remaining amount with his payment log.

4. Problem statement

Freelancers and independent contractors who manage expenses, invoices, income, and quarterly tax estimates in separate tools struggle to keep financial records consistent and timely. Duplicate payment entry, uncertain business-use deductions, and disconnected payment histories make it difficult to understand net earnings and plan estimated federal taxes. TaxWise Freelance will provide a single tenant-

isolated workflow for recording business activity, issuing invoices, calculating transparent planning estimates, and monitoring payment deadlines, with explicit assumptions and an estimates-only disclaimer. The MVP does not file returns, move money, connect bank accounts, or replace professional tax advice.

5. Five core MVP features and acceptance criteria

Feature	Acceptance criteria (Given / When / Then)	SRS mapping
Expense tracking and deduction categorisation	Given a $100 expense at 80% business use in a fully deductible category, when saved, then its deductible amount is $80. Given the same expense in Meals, then it is $40. Date, vendor, amount, category, notes, payment method, and business use persist; invalid percentages are rejected. Only the owner can read or change it.	FR-02
Income tracking	Given a valid manual receipt, when saved, then YTD income includes it once. Given an owned unpaid invoice, when marked paid, then exactly one income entry is created with its total and payment date; retries do not duplicate income.	FR-03, FR-05
Quarterly tax estimator	Given valid 2026 settings and records, when calculated, then the result exposes YTD and annualised profit, SE tax, half-SE deduction, QBI, standard deduction, bracket tax, total liability, payment target, amounts paid, and next due date. Safe harbour uses 100% or 110% of prior-year tax at the specified AGI boundary. All computation uses integer cents and the UI displays the disclaimer.	FR-04, FR-06
Invoicing	Given an owned client and valid line items, when an invoice is created, then its per-user number is unique and totals are calculated on the server. Draft, sent, paid, overdue, and void states follow controlled transitions; the printable view includes client, issuer, dates, items, and totals.	FR-05
Dashboard	Given records across several months, when opened, then KPIs and charts reconcile with income, expense, deduction, payment, and invoice records for the selected year. Outstanding invoices exclude paid/void records; upcoming deadlines use the selected tax year. Empty, loading, and error states are usable on mobile.	FR-07

The category catalogue includes Advertising, Car & truck, Contract labor, Insurance, Legal & professional, Office expense, Rent/lease, Supplies, Travel, Meals, Utilities, Software & subscriptions, Home office, Education, and Other. Categories map to Schedule C reporting lines or an explicitly documented subcategory/worksheet treatment; the app does not claim that a category alone establishes legal deductibility. Meals uses a 50% factor in the project model. Detailed home-office, mileage, depreciation, and eligibility calculations are outside scope.